

This API Endpoint Plan defines all routes required for RaceDay this ensures that organisers and participants can manage events, categories, enrolments and results in line with the approved functional requirements.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Register a new user account (Organiser or Participant) and create their account in the system.	Public	firstName, lastName, email, password, phoneNumber, role	201 Created - user with userId, sessionToken; 400 Bad Request - invalid input; 409 Conflict - email exists
POST	/api/auth/login	Authenticate a user by email and password, create server-side session, and return authenticated user details.	Public	email, password	200 OK - user object with userId, role, sessionToken; 401 Unauthorized - invalid credentials
GET	/api/profile	Retrieve the logged-in user's profile information including account details and role-specific data.	Logged-in User	None	200 OK - user profile; 401 Unauthorized
PUT	/api/profile	Update the logged-in user's profile information such as firstName, lastName, or phoneNumber.	Logged-in User	firstName, lastName, phoneNumber	200 OK - updated profile; 400 Bad Request; 401 Unauthorized
POST	/api/events	Create a new event record and assign it to the authenticated Organiser.	Event Organiser	name, description, date, location, startTime, routemap, eventType, registrationDeadline	201 Created - event with eventId; 400 Bad Request; 401 Unauthorized; 403 Forbidden
GET	/api/events	Retrieve a list of all available events in the system for browsing and entry.	Logged-in User	None	200 OK - array of events; 401 Unauthorized
GET	/api/events/{id}	Retrieve detailed information for a specific event including categories and route details.	Logged-in User	None	200 OK - event object; 401 Unauthorized; 404 Not Found
PUT	/api/events/{id}	Update an existing event's details; only the creating Organiser can modify.	Event Organiser	name, description, date, location, startTime, routemap, eventType, registrationDeadline	200 OK - updated event; 401 Unauthorized; 403 Forbidden; 404 Not Found
DELETE	/api/events/{id}	Delete an existing event and all associated categories, enrolments, and results.	Event Organiser	None	204 No Content; 401 Unauthorized; 403 Forbidden; 404 Not Found
POST	/api/events/{eventId}/categories	Create a new category or distance division for an event with entry fee and participant limits.	Event Organiser	categoryName, distance, categoryLevel, entryFee, maxParticipants	201 Created - category with categoryId; 400 Bad Request; 401 Unauthorized; 403 Forbidden; 404 Not Found

GET	/api/events/{eventId}/categories	Retrieve all categories available for a specific event to display entry options.	Logged-in User	None	200 OK - array of categories; 401 Unauthorized; 404 Not Found
POST	/api/enrolments	Register a Participant for a specific event in a selected category, creating participant_event record.	Participant	eventId, categoryID	201 Created - enrolment with participant_eventId; 400 Bad Request; 401 Unauthorized; 403 Forbidden; 404 Not Found; 409 Conflict - already enrolled or category full
GET	/api/events/{eventId}/enrolments	Retrieve all participants enrolled for a specific event with category and status.	Event Organiser	None	200 OK - array of enrolments; 401 Unauthorized; 403 Forbidden; 404 Not Found
POST	/api/results	Record the official result for a participant including race time, position, and status (Finished, DNF).	Event Organiser	participant_eventID, time, position, status	201 Created - result with resultId; 400 Bad Request; 401 Unauthorized; 403 Forbidden; 404 Not Found; 409 Conflict - result already exists
GET	/api/results/my	Retrieve all race results for the logged-in Participant across completed events with times and positions.	Participant	None	200 OK - array of results; 401 Unauthorized; 403 Forbidden
GET	/api/participants/me/performance	Retrieve aggregated performance history including best time, total races, best position, and average finishing position.	Participant	None	200 OK - performance object; 401 Unauthorized; 403 Forbidden

Notes – API Endpoint Plan

- **Registration handles role selection:** Organiser and Participant accounts are created through the same registration endpoint, with the selected role determining the user's access and profile.
- **Server-controlled values:** sessionToken, enrolment details such as registrationDate, registrationStatus and bib, and result identifiers should be generated by the system rather than manually entered by users.
- **Event ownership:** The authenticated Organiser is automatically associated with an event when it is created, preventing the client from assigning events to another organiser.
 - **Category capacity:** maxParticipants is included so the API can prevent registrations once a category reaches its limit.
- **Enrolment as the relationship:** PARTICIPANT_EVENT is created through the enrolment endpoint because it represents the Participant–Event relationship and stores category-specific registration information.
- **Result uniqueness:** Only one result should be recorded for each participant_event, preventing duplicate official results for the same entry.
- **Role-based access:** Organiser-only operations and Participant-only operations are restricted through authentication and role checks.
- **Performance data:** Performance history is treated as an aggregated view of a participant's completed race results rather than requiring the participant to manually enter performance information.
- **Ownership validation:** Event modification and enrolment access should verify that the authenticated Organiser is authorised for the relevant event.
- **Error handling:** 409 Conflict is used for business-rule conflicts such as duplicate enrolment, full categories and an existing result, rather than treating them as invalid requests.